

3D Vision: Project proposal

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1. Introduction

The main topic of this project is generative 3D object reconstruction from sparse multi-view RGB(+D) observations. Given a small set of images observing an object under limited viewpoints and occlusions, our goal is to reconstruct a complete object represented using 3D Gaussian Splatting (3DGS) [1].

We will build upon Object-X [2] which learns a structured latent representation (SLat) of geometry and appearance from multi-view evidence and decodes it into 3D Gaussian parameters for rendering. However, sparse viewpoints and occlusions make reconstruction ambiguous, as the observations may admit multiple geometrically plausible reconstructions. Our objective is to extend Object-X by strengthening its conditional latent space with a learned generative prior. We want to enable plausible completion of unobserved regions while remaining consistent with the observed views.

Key idea. We treat occlusion completion as region-specific latent inpainting. Object-X provides (i) an evidence-driven structured latent and (ii) a decoder from latent space to 3D Gaussians. We add a generative model (diffusion or flow) trained on Object-X latents. At inference, we clamp the observed part of the latent and sample only in unknown regions, optionally conditioned on category-level priors (class labels or text).

Expected outcome. Compared to the Object-X baseline, our approach should improve reconstruction under limited viewpoint and heavy occlusion, while preserving observed geometry and appearance and retaining efficient 3DGS decoding.

2. Proposed Method

2.1. Overview

Our pipeline has three modules:

1. **Structured Latent Representation Extraction** (from Object-X): sparse multi-view RGB(+D), masks, and poses are lifted into a voxel-aligned structured latent (SLat) z_{obs} .

2. **Generative latent completion (ours):** a conditional generative model completes the latent in regions not supported by the observations and/or the model.
3. **3DGS decoding (from Object-X):** the completed latent is decoded into 3D Gaussian parameters Θ for rendering.

2.2. Latent completion via region-specific inpainting

The generative module G is integrated between the Object-X latent extraction stage and the Gaussian decoding stage. It completes the partial latent z_{obs} using the instance masks M , producing a full latent z_{full} that is then decoded D into 3D Gaussians Θ and rendered R as the final output y . Generation can be optionally conditioned on a prior c (class label or text embedding).

Observation: $x \xrightarrow{E} (z_{\text{obs}}, M)$

Completion: $(z_{\text{obs}}, M) \xrightarrow{G} z_{\text{full}}$

Reconstruction: $z_{\text{full}} \xrightarrow{D} \Theta \xrightarrow{R} y(1)$

To ensure consistency with observed regions during generation, we enforce evidence preservation by clamping the observed voxels at each denoising step:

$$z_t \leftarrow M \odot z_{\text{obs}} + (1 - M) \odot z_t.$$

This constrains the model to modify only unknown regions while conditioning on the observed ones and the prior c .

2.3. Learning the object prior in Object-X latent space

Our approach learns the prior by training the generative model on fully observed objects z_{gt} . During training, we simulate partial observations by sub-sampling views and occluding parts of the object to obtain z_{obs} and the mask M , while using the complete latent z_{gt} as the target for completion. The model thus learns to generate the full latent z_{full} conditioned on the observed parts, the mask, and an optional prior c , enabling reconstruction even with limited evidence.

3. Algorithmic and Technical Challenges

1. **Challenge: Reliable observation mask M .** We have to determine which voxels are supported by evidence. With RGB-D, depth provides a strong signal. If we have RGB-only data, we rely heavily on multi-view projections and visibility. We validate the observation mask M with sanity checks. We visualized the observed region and compared the reconstructions with and without enforcing the observation constraint. The constraint should preserve consistency on the input views.
2. **Challenge: Data requirements** We need camera poses and object masks. So we should prioritize datasets where these are available (RGB-D sequences) or can be at least estimated reliably.
3. **Challenge: Evidence consistency under generation.** Generative completion could unintentionally modify parts of the object that are already supported by the input views. To avoid this, we kept the observed voxels fixed during sampling. So, only the unobserved region can change.
4. **Challenge: Compute requirements** As Object-X training will be expensive and requires GPUs we will need access to the cluster with sufficient GPU hours.
5. **Challenge: Ambiguity and multiple valid completions.** With strong occlusions, there is often no single correct shape for the unseen parts. So, basically several completions can fit the same evidence. Instead of forcing one deterministic prediction, we could generate a small set of candidates by sampling the generative model and then picking the one that best matches the input views.

4. Alternatives/Fallbacks

Even though diffusion models seem quite promising it might be too unstable or costly in terms of computation. For this reason, we could come back to the options below.

- **Conditional VAE:** cheaper stochastic completion than diffusion, still enabling multiple samples.
- **Flow-based generation:** fewer sampling steps than diffusion in some settings.

5. Implementation

5.1. Baselines

We could compare:

1. Object-X reconstruction from sparse views (no generative completion).
2. Deterministic latent completion.
3. Generative latent completion (diffusion or flow), with and without category conditioning.

5.2. Possible implementation flow

- **1. Step** Our goal is to get the Object-X pipeline running and transform the inputs to a rendered 3D Gaussian object using only the original Object-X reconstruction path. We could start with one object with a few views until the reconstruction looks correct. This will also help us debugging the voxelization, the feature lifting, decoding to Gaussian parameters, and rendering. Once this works, we can test on a small set of objects.
- **2. Step** We implement a simple and reliable way to compute the observation mask M to determine which voxels are supported by the input views. Then we create training pairs full objects with good view coverage and sparse versions using fewer views and simulated occlusions. Then we could implement a deterministic completion model eg. (3d CNN) that fills missing latent regions to simplify debugging also we can later use it as a Baseline to compare it to our diffusion model to argue improvements.
- **3. Step** Finally, we train the diffusion model in latent space to fill the unobserved region while keeping the observed region unchanged. Conceptually, this is latent inpainting: only voxels with $M = 0$ are generated, and voxels with $M = 1$ stay fixed. We could then test robustness by varying the number of views and the occlusion and compare against the deterministic baseline from step 2.
- **4. Step** We could add simple category-level conditioning to guide completions under heavy occlusion. Because completion can be ambiguous, we optionally sample several candidates and choose the one that best matches the input views.

5.3. Evaluation

We could evaluate:

- **Novel view synthesis:** PSNR, SSIM, LPIPS on held-out views as in the Object X [2].
- **Geometry quality:** if ground truth is available, evaluate geometric metrics or mesh-based distances after extracting a mesh from Gaussians.
- **Robustness:** performance versus number of views and occlusion severity.
- **Efficiency:** runtime and memory footprint.

References

- [1] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4), 2023. 1
- [2] Gaia Di Lorenzo, Federico Tombari, Marc Pollefeys, and Daniel Barath. Object-x: Learning to reconstruct multi-modal 3d object representations, 2025. 1, 2